

Review

The dynamics of peri-urban expansion in sub-saharan africa: implications for sustainable development in Nigeria and Ghana

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Abstract

Peri-urban land use is diverse and rapidly expanding in sub-Saharan Africa (SSA). It is characterised by haphazard development and unplanned land, which has become a significant global preoccupation due to environmental challenges. The issue of PUAs in sub-Saharan Africa has dramatically concerned many scholars and stakeholders, including city authorities and professionals in urban planning and management. This study evaluates the differences and similarities in the trends and consequences of the spread of SSA peri-urban areas (PUAs), focusing on Nigeria and Ghana. The urban–rural linkages and flows provided the theoretical background for the study. The study solely relied on secondary data through the desktop methodology to obtain specific data on development, sustainable development, peri-urban areas (PUAs), urbanisation, and urban expansion in sub-Saharan Africa, particularly Nigeria and Ghana, from databases such as Google, Google Scholar, ResearchGate, and Mendeley. Other sources consulted intermittently for necessary information were relevant books, conferences, working papers, and reports from global organisations. The study found that the expansion of PUAs in Nigeria and Ghana is not at the same level due to different population sizes. The study also found that the reasons for PUA expansion in Nigeria are not similar to those in Ghana. To enhance sustainable growth, the study recommended that the government or city authorities take over the lands on the urban fringes and design a plan that includes the road networks and other specified purposes for the land. This will guide against the spread of disorganised and unplanned neighbourhoods currently experienced in both countries.

Keywords Sustainable · Peri-urban · Sub-Sahara · Dynamics

1 Introduction

The term ‘development’ is expansive; hence, the academic literature is rich in studies concerning it and its affiliated subject matters [1–3]. Its broad spectrum means many definitions expressed from different scholarly points of view and ideologies exist, making it difficult to have a consensus definition [2]. Notwithstanding, this study adopted some functional definitions that provide the necessary framework for this study. For instance, Peet and Hartwick [4], considered development “the founding belief of the modern world connoting progress, modernity, and democratic values”. Furthermore, development has also been used in various contexts across the globe to describe the status of countries relative to economic, socio-cultural, and environmental factors and how these factors could be deployed in the transforming local and national economies, societies, and the environments [1, 2]. The emphasis in the preceding description that development is not dependent on a singular factor is consistent with the opinion of Todaro and Smith [5], that development is a ‘multidimensional process which involves critical changes in social structures, popular attitudes, and national institutions,

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as well as boosting economic growth, reduction of inequality and poverty eradication.' Therefore, it can be implied that development is a reflective mirror that shows the varying incidence of growth using multidimensional yardsticks, hence the tradition of describing countries as developed, developing, or underdeveloped.

The African continent (especially its sub-Saharan parts) is vastly underdeveloped relative to the other continents; thus, it has been repeatedly described as a continent in a developmental dilemma [2, 3]. According to the United Nations [6], 'sub-Saharan Africa (Fig. 1) are the African countries located south of the Sahara, excluding North Africa with the Sahel as the Sahara's transitional zone and the Sudan region's tropical savanna. This means sub-Saharan African (SSA) countries occupy the continent's eastern, southern, central, and western parts, and manifest features of underdevelopment. This is bewildering given the vast deposits of biological and mineral resources of immense monetary worth plus the enormous human resources that abound in this region. Furthermore, eight of the thirty-six global biological diversity hot spots are located in this part of the African continent, at the same time, the continent is also home to an estimated 50,000 known species of plants, 1,500 bird species and 1,000 mammals [7, 8].

Academic researchers have, over time, offered various explanations for the intense state of underdevelopment of the continent amid the undisputable layers of invaluable wealth scattered across its length and breadth. Some of the most typical explanations on this subject matter have been based on the theories of development and underdevelopment, such as the modernization theory, the dependency school theory, the third-world analysis theory, and

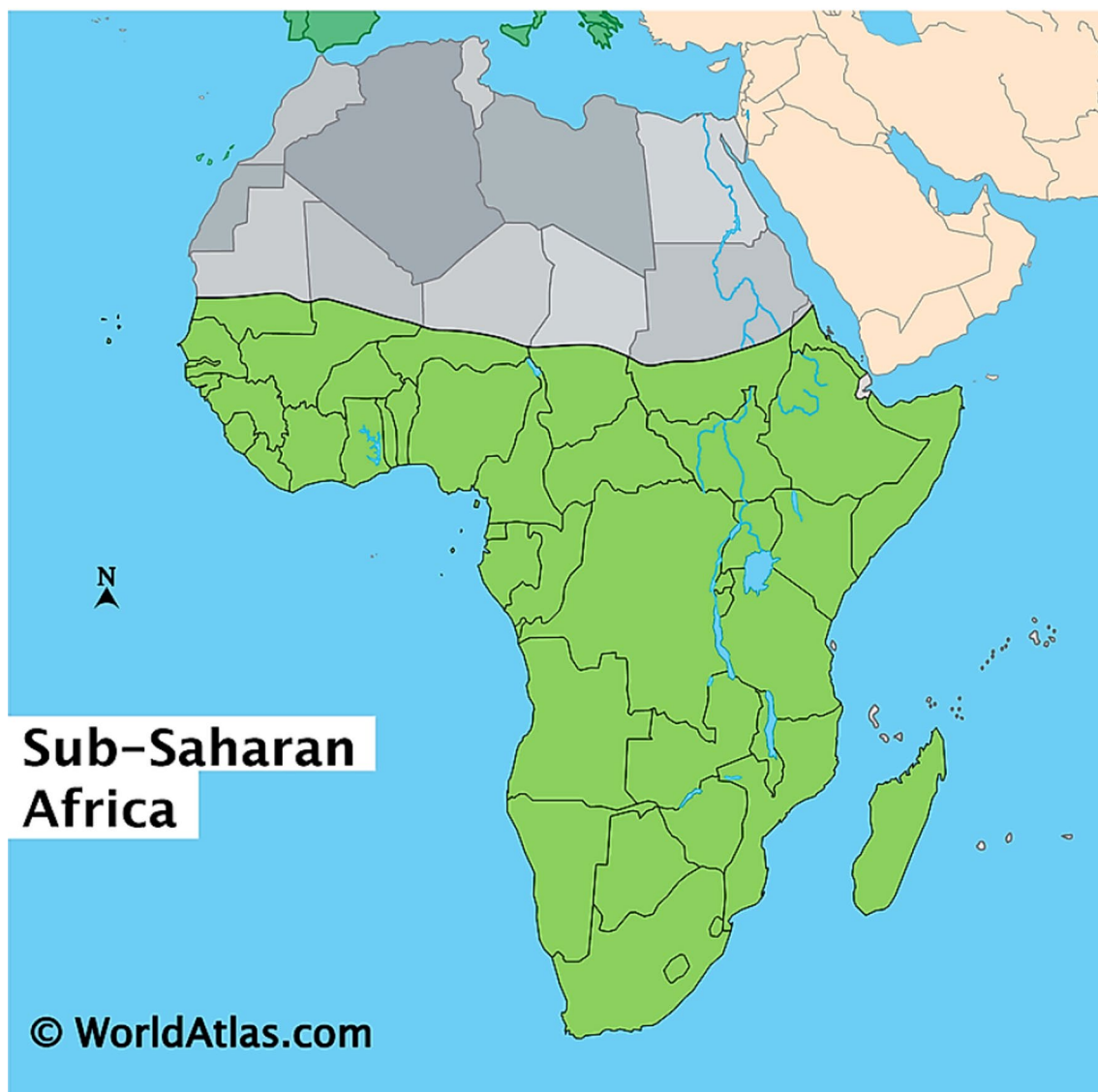


Fig. 1 Sub-Saharan Africa. Source: WorldAtlas.com <https://www.worldatlas.com/regions/sub-saharan-africa.html>, United Nations [6]

the globalization perspectives [2, 9]. Their submission is that underdevelopment in SSA is a function of certain factors some of which are disease susceptibility and geographical location, colonialism and slave trade, capitalism and adjoining policies as well as weak African institutions [2]. The consequences of these drivers manifest as low GDP/per capita income, corruption, unemployment, a high population of rural dwellers engaged in agriculture, a high poverty rate, political instability and a poorly developed political structure, use of inferior and low-quality technology, lack of basic infrastructural facilities, a weak educational system, and a high death rate amongst several other factors that epitomize the continent's state of being [3].

The aforementioned debilitating factors have contributed to the widening the development gap between sub-Saharan African countries to countries from other climes. Their cumulative effects are commonly seen using robust development indicators like the Human Development Index (HDI). For instance, the average HDI score for the African continent estimated from the year 2000 to 2022 revealed that more countries have always had scores (0.43–0.55) that correspond to the low human development tier of the HDI [10, 11]. This means that African states averagely have a low life expectancy, the state of education is below par and economic growth is defective. The HDI scores when explored based on individual countries, revealed that only one African country (Mauritius) has a very high HDI (0.8–1.0), 3 others have high HDI scores (0.7–0.79), 16 African countries have medium HDI (0.55–0.7), and the final group with 26 countries has low HDI (below 0.55) [11].

Peri-urban land use is diverse and very dynamic in sub-Saharan Africa. The dynamism results from the continuous unmonitored land use resulting from the uncontrolled growth and spread of Africa urban areas. Peri urban areas (PUAs) worldly are susceptible to urban sprawling due to their proximity to urban areas [12–14]. Thus, the unprecedented population growth in urban areas across the globe has been cited as the primary driver of urban sprawling [12]. This claim can be substantiated by extant statistics, which put the global urban population growth in the last seven decades (1950–2020) at 56%, with further prediction suggesting that a further increase to 68% is expected by 2050 [15]. Relative to the African continent, the urban population is expected to triple from 395 million in 2010 to 1.339 billion in 2050 [15]. However, the sad reality is that most urban areas in sub-Saharan Africa are poorly equipped to handle either the current growth or the forecasted population influx because most of them are saturated and the existing infrastructures are already overwhelmed; thus, the situation of sprawling into adjoining unplanned areas (peri-urban areas) and the subsequent emergence of many informal and illegal settlements which accommodate the new urban residents is likely to continue unabated [12–14]. The multidimensional consequences of urban sprawling on sub-Saharan Africa's PUAs manifest in multiple ways that collectively mold the general environmental conditions of the area [16, 17, 125, 126, 128–134].

Furthermore, the haphazard spatial patterns manifested by urban sprawl in PUAs not only make the management of the PUAs difficult but also portends danger to sustainable development [16, 18]. An indication of the intensity of urban sprawling in PUAs was captured by Naab et al. [19] where they stated that the fast rate of urban development in PUAs of sub-Saharan Africa had altered the land use dynamics by causing multiple land use conversions such that an estimated 14 million hectares of land was claimed to have been lost between 1990 and 2020 at the rate of 475,000 hectares per annum. In addition, there have been a plethora of studies [3, 7, 9, 13, 14, 16, 19, 21, 30, 34, 53, 54, 62, 70, 80, 84, 87–91, 94, 95, 98, 99, 102, 104, 107, 108, 113–115, 118, 126, 127, 130] on peri-urban areas globally, continentally, in Nigeria, Ghana, many other sub-saharan African countries. However, this study is the first to evaluate and compare the nature of peri-urban in these two sub-saharan African countries (Nigeria and Ghana) to understand the differences and similarities in the haphazard development and unplanned land use that characterized the peri-urban areas of sub-Saharan Africa. Furthermore, although urbanisation is a positive development, it still negatively affects peri-urbanism. Based on the preceding, this study evaluates the trends and consequences in peri-urban areas (PUAs) in sub-Saharan Africa (SSA), focusing on Nigeria and Ghana.

One of the salient takeaways from the preceding is that sub-Saharan Africa's underdevelopment is multifaceted, and as such, the development process may also emanate from multiple sources. Furthermore, development, as previously stated, is tied to economic, socio-cultural and environmental factors [1, 2]; hence, the journey of sub-Saharan Africa out of the doldrums of underdevelopment lies in the domain of these factors. This study, therefore, aims to evaluate the differences and similarities in the trends and consequences of the spread of peri-urban areas (PUAs) of SSA with a focus on Nigeria and Ghana.

The specific objectives of the study are to assess the expansion of PUAs in Nigeria and Ghana; ascertain the differences and similarities in the process of PUA expansion in Nigeria and Ghana; identify the consequences of PUA expansion in Nigeria and Ghana; recommend practical and straight-forward remedies for PUA planning in both countries. The study's

research questions sought to know what the expansion processes of PUA look like in Nigeria and Ghana? Are there similarities in the trends and consequences of PUA expansion in Nigeria and Ghana?

The study is divided into eight sections. The first section provided the introduction of the study, which included the background of the study where necessary background information is provided. Also, the statement of the problem was discussed to identify the essential gap that prompted this study. The significance and justification of the study are also discussed in the introduction section. The literature review and the necessary theories for the study are also discussed in this section. The second section addressed the methodology. The third section presented the study area, including a map to showcase the area known as sub-Saharan Africa. The fourth section presented the results, which provided an in-depth discussion on issues about peri-urban constraints and sustainable development, trends and consequences of PUAs in sub-Saharan Africa, and case studies of Ghana and Nigeria where the differences and similarities in PUAs expansion are fully presented. Section five provided a discussion of the findings and insights. Section six discussed the recommendations and implications of the study. The overall contribution of the study is discussed in the seventh section, while section eight presents the study's limitations.

1.1 Research significance and justification

The significance of this study further enhances the understanding of the differences and similarities of PUAs' expansion. This understanding is expected to strengthen the efforts to address the unplanned expansion of PUAs, particularly in SSA. The study's justification lies in its potential to reveal the similarities and differences in the process of PUA expansion in the selected countries. This would positively impact the knowledge of the extent of PUA expansion and challenge the scholars and stakeholders to find ways to deal with it. The study's approach in comparing the expansion of PUAs in two different countries in SSA is to bring to the fore the similarities and differences in how PUAs are expanding in this region. It is one critical step in finding ways to address the spread of PUAs in SSA.

1.2 Review of literature

The review of literature in this study is presented in themes. The themes included the trends in the development of peri-urban areas and the consequences of the spread of peri-urban regions. Peri-urban is subject to varying definitions among different authors. The term "peri-urban" is otherwise known as "urban sprawl". The term "sprawl" encompasses several scenarios that exhibit patterns, depict a particular process, or arise from specific factors. According to OECD [20], peri-urban is a growth pattern distinguished by a low population density. Peri-urban is often described by many authors using various terms such as low-density, scattered, and discontinuous urban development patterns. It refers to the physical expansion of urban areas into rural and natural areas adjacent to urban centres. This expansion often involves leapfrogging away from denser areas and transforming open, undeveloped land into single-family residential homes. Urban sprawl can occur as unplanned and planned growth, resulting in sporadic settlements. These descriptions have been provided by researchers such as Gomez-Antonio, Hortas-Rico, and Li [21], Egidi, Cividino, Vinci, Sateriano, and Salvia [22], and Aurambout et al. [23]. Egidi et al. [22] assert that urban sprawl is a contentious phenomenon that leads to the spatial reconfiguration of cities in both developed and developing nations. Peri-urban, similar to urbanisation, entails the migration of individuals from rural to urban areas, resulting in a rise in population in the receiving regions. In this study, the researchers defined urban sprawl as the deliberate or unintentional growth of residential housing into agricultural and municipal areas. Parnell and Pieterse [119] interrogated the paradox that although the African continent is roughly forty percent urbanized, comprising over 400 million inhabitants, Africa is still primarily viewed as rural. Parnell and Pieterse [119] identifies some significant challenges facing Africa urbanization. Also, Sumardjo et al. [127] examined peri-urban community resilience in facing global climate-change impacts toward sustainable development. The study identified research gaps that are essential to policy relevance in SSA. Addressing these gaps is imperative to harness the economic advantages of rapid urbanisation in a way that supports rural areas and promotes sustainable development. These challenges includes inequalities, peri-urban growth, under-employment, outdated colonial urban management schemes, and competing sources of power in urban settlements. They concluded, however, that the future of Africa is in the growth of urban and that the stakes are too high to ignore the complexity and importance of African cities of all sizes.

Peri-urbanization is the progressive increase in the urban population in the city's fringes, leading to significant impacts on the economy, political landscape, and cultural significance of cities compare to rural areas [24, 25]. In her study, Makowska-Iskierka [25] characterized urbanization as a multifaceted phenomenon that entails a series of stages, resulting

in the irreversible transformation of rural regions into urban areas. Urbanisation, as commonly defined in scholarly papers, refers to the movement of people from rural areas to urban settlements [24, 26]. Urbanisation in wealthy countries began due to industrialization. Still, in many developing countries it is driven by high population growth and migration from rural to urban areas, leading to urban expansion [27]. It is evident from the aforementioned that urbanisation mainly arises from industry, the natural growth of population, and rural–urban movement. Idowu, Bako and Aduloju [120] examined the pattern in the peri-urban region's change between 1972 and 2015 and discovered that Minna's built-up area had increased by nearly 2000%. With an increasing pace of uncoordinated and unplanned development, the study further suggested that peri-urban areas of Minna by 2015 had grown roughly thirty-two times their size in 1972; thirteen times their size in 1979; three times their size in 1986 and 1996, respectively; and twice their size in 2006. It concluded that strategies guiding Minna peri-urban areas should not let chance or the interaction of economic variables define their planning and development; instead, they should be guided by ideas ensuring orderly growth and development. According to the Tagliarino et al., [28], it is estimated that by 2035, urban areas will be home to half of the inhabitants of the African continent. The impacts of urbanisation, similar to those of urban sprawl, can have both detrimental and beneficial consequences. Among the negative effects of urbanisation are sprawl as a result of unplanned and uncontrolled city growth, pollution resulting from overcrowding, which leads to public health issues, increased unemployment, shortage of affordable residential housing, transportation challenges for commuters, inadequate infrastructure and poor service delivery (water, solid waste management, and water waste treatment), cities becoming too expensive for low-income earners, encroachment into agricultural land, development of slums (squatters), increased urban poverty, and crime [24, 26, 29]. The positive effects of urbanisation simply represent the benefits of urbanisation, which include economic improvement, commercial growth of activities (growth in trade and tourism), social and cultural integration (people of different ethnicities, religions, and income levels can stay and work together), the provision of better services and transportation networks, and efficient services [24, 26]. The report from the Thabit et al., [81] provides additional evidence to support the positive impacts of urbanisation. It states that when urbanisation is appropriately planned and managed, it can have economic benefits and stimulate innovation and productivity, thereby contributing to the developing national and regional areas and helping to alleviate poverty.

Furthermore, the Dyachia et al. [135] study looks at the missing link between land use planning and urban growth and its implications for environmental sustainability in a selected sub-Saharan African city of Kaduna, Nigeria. The study assessed urban growth from historical GIS data of the town to evaluate the urban expansion and establish a relationship between carbon emission and traffic volume in the city. The study observed that the lack of coordination between land use planning and urban growth has resulted in environmental ills within the Kaduna metropolis. The study's findings further show a strong connection between emissions, traffic volume, and the decline of green areas in the city.

1.3 Theoretical framework

This study is guided by two different theories namely, the sustainable development concept and the urban–rural linkages and flows theory. The theories are discussed in a way that provides an understanding of the concepts related to the argument advanced in this study.

1.3.1 The concept of sustainable development and peri-urban areas

Sustainable development (SD) is an agenda to promote the judicious use of available resources to satisfy human needs and enhance development in the present and future [35, 36]. This narrative suggests that sustainable development ensures the responsible usage of all categories of resources to preserve their longevity and capacity to meet human needs. Hence, the concept is most needful in regions like the PUAs of sub-Saharan Africa where the diversity of human-induced mishaps has given rise to unsustainable certain prevalent features. Some of these features include the dissipation of agricultural land and depletion of natural resources (like forest and high-biodiversity wetlands), extreme pollution, poor sanitation, abysmal waste management, absence of basic infrastructures, land and water resources strain, defective layout plans, land tenure debacles, and land ownership contention [16, 37–40]. Furthermore, the PUAs are characterized by high population growth induced by rural migration, illegal and fraudulent developments, socio-cultural diversity, squatter settlements, and areas devoid of urban planning [16, 41]. The SD concept has evolved into a comprehensive tool that hinges on three fundamental pillars, namely the environmental, economic, and social [42–45]. Hence, the prevalence of SD models that highlight and encapsulate the interplay between and among these pillars through the provision of the spine needed to understand and address complex issues like those that typified the PUAs in sub-Saharan Africa [42–45].



Fig. 2 Munasinghe's Triangle. Source: Pearce [42]

Munasinghe's triangle is one such model that attempts to find solutions to sustainability by harmonizing the benefits and side effects of the three pillars [42]. This means the model tries to balance the pros and cons of the respective pillars to generate a sphere of sustainability. Sub-Saharan Africa's PUAs would benefit from a reasonable framework that could bring about some semblance of stability to its domain; the whole essence would be to create a zone of sustainability that acknowledges and cater to the linkages of the environment, economic and socio-cultural elements in the domain [46].

The stipulations of Munasinghe's triangle in application to the PUAs' environment is to ensure that all the living systems (forests, agricultural lands, shrub-lands, rivers, streams, ponds, wetlands, and others) in the environment are deployed in ways that are capable of maintaining their viability and health using measures that are multi-dimensional, thorough, effective, resilient, robust, and efficient [47]. Relative to economic sustainability, the ideology revolves around maximizing income flow from existing capital without compromising the usability and health of the assets through excessive consumption and overexploitation [48]. Social sustainability focuses on strengthening human capital (like education and skills) and socio-cultural values (social relationships, communal practices, customs, norms and values, and other cultural nuances) that are peculiar to the residents and held in high esteem [42]. Areas of benefits from Munasinghe's triangle to peri-urban environmental issues include the regulation of pollution, deforestation, land clearing, land use conversion, and other such issues that constitute ecological degradation. This further connotes that practices that would reinforce the resilience of the ecosystems within the environment both at larger and smaller spatial scales can be promoted if environmental sustainability is achieved [49]. The target of peri-urban economic sustainability should be what Pearce [42] called the effort to attain economic efficiency through optimal consumption and production using peri-urban assets. The existing peri-urban capitals are the living, environmental components and social assets from which production and consumption evolve. Thus, it is imperative to use these resources judiciously by ensuring their consumption will not affect the well-being of the dependents or plunge them into poverty. Identifying the ecological and social capital categories that require maintenance and management, because failure may induce calamitous consequences [50, 51]. Resource harvest should not overshadow its replenishment rate nor alter the ecosystem's natural balance [51]. Furthermore, while resources like land (which is a foremost economic resource) should be used to offer more economic advantage, it is also imperative that the uses are not crooked; instead, they should be balanced. Furthermore, the need to strengthen peri-urban human capital and socio-cultural values cannot be overemphasized because it is vital to the sustainability efforts directed at the other available peri-urban capitals. Hence, continuous individual and communal education, establishing active, and focused institutions, and other measures that could aid the preservation of cultural capital and its diversity in the PUAs would be beneficial [42]. Figure 2 shows a schema of Munasinghe's triangle.

1.3.2 Urban–rural linkages and flows theory

The implication of urban–rural linkages and flows theory to this study refers to the systemic socioeconomic problems and inequality in the urban centres, which encourages a mass movement of people from urban areas to the fringes. This act contributed to the spread of peri-urban development because many people could not afford the basic necessities of life in urban centres. Typically, peri-urban zones are once rural areas that have become urbanised and are now dependent

on the resources of their environment. Planning and policy neglect of peri-urban areas has exacerbated inequality in social, environmental, economic, and health outcomes [124]. The urban–rural linkages and flows theory was adopted in this study. Using the overarching framework of dependence theory, Unwin [125] examined the urban–rural dynamics in developing nations. The ideological decision that Potter uses to support his last piece does not provide much beyond outlining a socialist framework to study urban–rural linkages in an abstract form. One theory holds that a country's planning issues stem from its settlement pattern, reflecting more systemic socioeconomic problems and inequality. To address these issues, it is necessary to consider "more fundamental socio-political issues that are of crucial significance, not least the mode of production and the development path that has been adopted in particular countries," rather than focusing on specific urban solutions such as growth poles or new towns.

2 Methodology

This study evaluates the trends and consequences of PUAs in sub-saharan Africa focusing on Nigeria and Ghana. The study adopted the desktop methodology for the data collection in this research review, which means the data in use were secondary data sourced from different outlets (see Table 1). The justification for desktop methodology is that global agencies process reliable data for the study's objectivity. Relevant keywords about the review topic were fed into selected online databases to obtain data on them. The online databases provide easy access to existing literature especially peer-reviewed journals on varieties of subjects including this review. Therefore, databases like Google, Google Scholar, ResearchGate, and Mendeley were consulted intermittently for necessary information. Additionally, the databases provide easy access to relevant books, conference, and working papers, as well as reports from different global organizations like the United Nations (UN), UNDP, UNICEF, UNDESA, the World Bank, the African Development Bank (AfDB), the Infrastructure Consortium of Africa (ICA), IBRD, UNFPA, NEPAD, and others. Some specific keywords used for data search include development, sustainable development, peri-urban areas (PUAs), urbanization, urban expansion, sub-Saharan Africa, and several others. Details are provided in Table 1.

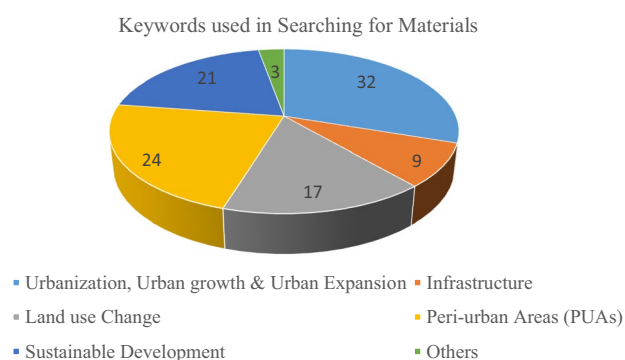
Table 1 and Fig. 3 summary all the materials downloaded, read, and used in the review between 10th September 2022 and 30th November 2022. While Table 1 summarises the data sources, which clearly shows that journals (69.9%) and reports (16.2%) were the predominant sources consulted, Fig. 3 gives the analysis based on the keywords used in searching for relevant materials. Furthermore, Fig. 3 revealed that the keywords used in the search were urbanization, urban growth, and urban expansion (30%), PUAs (23%), sustainable development (20%), land-use change (16%), infrastructure (8%), and others.

Table 1 Source of Research Data

Sources of research materials/data	Number of materials	Percentage of materials (%)
Journals	95	69.9
Books	8	6.6
Reports	22	16.2
Conference Papers	2	1.5
Internet Materials/Magazines	4	2.9
Working Paper	2	1.5
Unpublished Dissertations	2	1.5
TOTAL	136	100.0

Source: Authors, 2023

Fig. 3 Distribution of Academic Papers used by Keywords. Source: Authors, 2023



3 Study area

Sub-Saharan Africa (SSA) or Sub-Saharan is the area and regions of the African continent that lie south of the Sahara (See Fig. 1). These areas are Central Africa, East Africa, Southern Africa, and West Africa. The only regions of Africa excluded in this description are North Africa and the northern part of West Africa. The two countries, Nigeria and Ghana, which are the focus of this study, sit in Western Africa, which is situated in SSA.

4 Results

4.1 Peri-Urban constraints and sustainable development

Various studies have been conducted to investigate issues arising as a consequence of peri-urbanization across the world [18, 52]. Challenges of varying magnitude from those tied to the environment, to economic, sociocultural factors, and political factors have been identified [53, 54]. This is primarily because the urbanization process, which kick start peri-urbanization is a process that consume space and is spread to different land uses [55]. However, identifying the chief drivers of PU underdevelopment means pinpointing the indicators of sustainable development lacking in PUAs because these indicators are sacrosanct for development to occur and be sustained. An indicator according to the OECD [56], is “a parameter or a value derived from one or more parameters, that indicate supplies information, describes the state of a phenomenon, environment, or zone, with a significance which extends beyond that which is directly associated with the parameter’s value”. The variety of functions captured for an indicator in the OECD definition aligns with the opinion of Rey-Valette and Chia [57] that many functions characterized an indicator. Thus, Balestrat [58] also stated that indicators are communication tool which means they are decision-making tool. This explains why indicators are deployed as tools for gauging development; they communicate fundamental information to decision-makers. Table 1 gives a synopsis of the core sustainable development indicators that should be available in any domain of human habitation for development to be a surety.

As shown in Table 2, the respective pillar of SD has multiple themes (which highlight the multidimensional nature of sustainable development) that are attached to them; these themes have measurable sub-themes (check supplementary material) as well as guidelines and methodologies for their estimation [15, 59]. However, the list is too detailed, making national testing using all the indicators nearly impossible. Thus, experts have recommended that each country select indicators from the exhaustive list that would best suit the issues of their respective sustainable development policy [15]. Therefore, this broad list represents the general list from which most requirements for gauging urban growth and development are gleaned [52, 60]. For instance, Bloch et al., [60] gave the following twelve (12) indicators as the criteria for measuring the performance of Glasgow city, these include population, economic participation, poverty, health, social capital, environment, transport, education, the safety of local communities, lifestyle, cultural vitality, and mind-set. Each of these indicators has sub-indicators that give clarity to each indicator.

Therefore, the non-availability of these indicators will stall sustainable development [52, 59]. So, considering the peri-urbanization process from the perspective of indicators, some of the major constraints of SD in sub-Saharan PUAs are categorized as follows:

Table 2 Core Sustainable Development Indicators

Social	Environmental
Education	Freshwater/groundwater
Employment	Agriculture/secure food supply
Health/water supply/sanitation	Urban
Housing	Coastal Zone
Welfare and quality of life	Marine environment/coral reef protection
Cultural heritage	Fisheries
Poverty/Income distribution	Biodiversity/biotechnology
Crime	Sustainable forest management
Population	Air pollution and ozone depletion
Social and ethical values	Global climate change/sea level rise
Role of women	Sustainable use of natural resources
Access to land and resources	Sustainable tourism
Community structure	Restricted carrying capacity
Equity/social exclusion	Land use change
Economic	Institutional
Economic dependency/Indebtedness/ODA	Integrated decision-making
Energy	Capacity building
Consumption and production patterns	Science and technology
Waste management	Public awareness and information
Transportation	International conventions and cooperation
Mining	Governance/role of civic society
Economic structure and development	Institutional and legislative frameworks
Trade	Disaster preparedness
Productivity	Public participation

Source: OECD [56]

- Accessibility & Infrastructural Constraints
- Land use conversion
- Ecosystem degradation
- Institutional Debacles

4.1.1 Infrastructural constraints

Peri-urban areas are areas of multiple inequalities, with infrastructural debacles being one of the primary challenges. This is especially true for sub-Saharan PUAs where infrastructures access is highly challenging [61]. The domain lacks some basic infrastructures for human well-being such as water, electricity, transportation system, communication networks, sewage, healthcare facilities, housing, and school systems [23, 62]. For instance, Calderon et al. [61] reported that access to water in rural areas of SSA stands at 67% in 2015 while it was 85% in other developing regions of different continents. Therefore, the importance of infrastructures to the development of any domain cannot be overemphasized; hence infrastructures have been described as a precondition for sustainable development [61, 62]. This is because the availability of infrastructure and its quality determines the success or failure of many other activities within any domain of human habitation.

Research findings have confirmed that the infrastructural stock of sub-Saharan Africa is grossly inadequate [61, 63, 64]. Most reports like those of the World Bank, AFDB, and the Infrastructure Consortium of Africa (ICA), put the prime infrastructures that serve the populace into four categories, namely transport, energy, water (and sanitation), as well as Information Communication Technology (ICT) [61, 65]. According to the African Development Bank (AFDB) [66], the annual estimate of these infrastructural needs of the continent stands between \$130–170 billion, but, most African countries could not completely meet this financial obligation, thus resulting in a cumulative financing gap of \$68–108

Table 3 Infrastructural Financing Gap for Africa

Sector	Annual Financing Need	2016–2018 Commitment Average	Financing Gap
Transport	35–47	31	4–16
Water	56–66	13	43–53
Energy	35–50	30	5–20
ICT	4–7	4	0–3
Total	130–170	78	52–92

Source ICA [65]; AfDB [66]

billion [66] as presented in Table 3. However, this reported gap in infrastructure financing is new because the previously reported gap given by the World Bank was \$31 billion [67]. This means that the financial requirements for boosting SSA infrastructures have increased with every urban population surge. This can be attributed to the rapidity of the urbanization process in SSA with the region expected to experience a 300% increase in urban population and induce urban sprawling between 2010 and 2050 [68].

McConville and Wittgren [69] summarized the situation of PUAs infrastructure to be such that the population growth rate outpaces the rate of infrastructural development, which means a sizable part of the population will not have access to infrastructures. This unwanted situation should ideally be managed and possibly corrected by governmental institutions. However, their inability to control the infrastructure situation in the urban areas suggests that sustainable development in PUAs would remain fuzzy. The financing gap in Table 2 indicates the financial gulf that should be submerged for Africa infrastructural challenges (especially for water and energy) to be checkmated.

4.1.2 Land use conversion

SSA's peri-urban areas are characterized by poor land use planning; thus spates of land use conversion are commonplace within the PUA domain [70]. According to Natural Water Retention Measure, NWRM [71] land use conversion involves large-scale geographic change. It is a change from the prime use of land to another use that may induce changes in its vegetation cover and other features. Globally, forests and agricultural lands within the PUAs are frequently converted to various land uses, including housing projects and building constructions, highways, roads, and bridge constructions, and building of industrial estates amongst others [72]. Although, some studies have argued that these conversions are inevitable [73], their magnitude in PUAs is often sizable; thus, the attendant consequences are often harmful to the affected ecosystems [71]. For instance, Rawat & Agarwal [74] cited land use conversion as a danger to biological diversity as it induces habitat alteration, overexploitation, climate change, impairs water quality, etc.

Extant literature has also highlighted land use conversion as an aftermaths of urbanization and urban growth in the PUAs of SSA [75]. Urban population statistics for SSA have revealed that the population of urban dwellers has increased by a cumulative percentage of 41.83% in the last 61 years (1960–2021). In West Africa, Herrmann et al. [76], Asenso, et al., [77] highlighted the extent of land use and land cover change when they pinpointed that the areas inhabited by humans between 1975 and 2013 (38 years) increased by over 200% from 493,000km² to 1,121,000km². These land use changes would involve land use conversion. For example, forest areas may be cleared for several unplanned uses like agriculture, road construction, and building of residencies, and factories. Therefore, these statistics are indications that unplanned land use changes like land use conversion would continuously hamper the peri-urban areas which may pose a considerable challenge to sustainable development.

4.1.3 Ecosystem degradation

The volume of environmental and ecosystem-impairing anthropogenic activities in sub-Saharan Africa's peri-urban areas is enormous [37]. Activities like deforestation, land clearing, overgrazing, bush burning, mining and quarrying, wetland conversion, excessive logging and firewood harvesting are notable ecosystem stressors that are commonplace in these areas and capable of stalling sustainable development [37]. These degrading activities affect the composition of the ecosystem like water, air and soil, through endless exploitation of the existing resources at a rate that outpaces the ecosystem replenishment process, thereby creating an imbalance that could trigger biodiversity displacement and extinction, climate change, soil erosion, pollution, and water contamination [37, 77].

According to the United Nations Environment Programme (UNEP), SSA has arable land, water, oil, natural gas, forests and wildlife. However, the prevalence of ecosystem degradation in locations like the peri-urban areas suggests that these natural capitals are consumed unsustainably, threatening people's sustainable livelihood, health, economic security, and the global environment [78]. For instance, the World Bank [78] highlighted that 65% of SSA's soil has become degraded, leaving just 20% of arable land for planting; furthermore, approximately 50 million acres of forest have been lost since 1980; the burden of disease (like malaria, diarrhea and respiratory infections) which emanates from the poor state of the environment also means that people from the region are exposed to continuous health risk [78]. The proceeding implies that ecosystem degradation worsens people's vulnerability as it could cripple them environmentally and economically and therefore constitute a barrier to sustainable development.

4.1.4 Institutional debacle

Neuwirth [79] stated that the population of people living in informal settlements across the globe is close to one billion. This shows that the peri-urban phenomenon is global; however, the intensity and severity of the challenges that come with it are astronomical in developing regions like SSA and South Asia [80]. Relative to SSA, about 62% of the people live in informal settlements, primarily unplanned areas situated at the fringes of urban areas; likewise, 43% of people in South Asia also live in similar unplanned settlements, thus it could be concluded that informal PUA development would be the predominant form of urban growth in most developing countries [82]. Therefore, the lack of basic plans for the settlements and communities emanating from these localities is worrisome.

The prime issues which range from a lack of infrastructures (roads, water, electricity, energy, etc.), profound degradation and pollution, series of land use changes, and conversion, ecosystem conservation, economic growth, and social equity, highlight the institutional neglect that PUAs have come to be known for [80, 83]. The inability of the government to properly manage the situation in the PUAs has been traced to a variety of issues, primary amongst which are financial inadequacy, lack of adequate administrative tools (like land sub-division maps and proper boundary delineation), lack of effective initiatives (like stake-holders partnering or Public–Private Partnership), corruption and compromise of standards, and other such issues [80, 83, 84].

4.2 Trends and consequences of PUAs in Sub-Saharan Africa

The rapid spread of sub-Saharan Africa's peri-urban areas has its attendant positive or negative consequences. Peri-urban areas are areas between town/city and countryside/rural areas. It could also be called the rural–urban transition zone where, urban and rural activities and functions mix and often clash. Peri-urban areas have continued to be an area of interest to many scholars mainly because of the rapid and increased spread in many cities in the sub-Saharan Africa. According to UNDESA [85], it was claimed that Africa and Asia accounted for about 90% of global urban population growth, which is primarily caused by urban sprawl. In contrast, UNECA [86] submitted that East Africa is the least urbanised sub-region in sub-Saharan Africa but urbanising the fastest. Southern Africa, the most urbanised, is urbanising at the slowest rate. The increasing spread of peri-urban areas is quickly becoming visible in many sub-Saharan Africa cities and towns. For instance, the most significant cities and towns in sub-Saharan Africa have their fringes growing into large-scale urban centres, usually referred to as peri-urban areas that are as big as the size of the urban area. Several reasons have been recognized as the primary causes responsible for the increased spread of peri-urban areas in sub-Saharan Africa's cities. Some of the causes are the increasing number of immigrants searching for jobs in the cities/towns, which in turn led to population growth leading to increasing demand for accommodation, the need for commercial and recreational land uses, and the need to engage in agricultural activities to feed the cities [87]. Lawanson, Yadua [39] and Acheampong & Anokye [88] submitted that spread of PUAs is motivated by the fact that affordable rents are much available in the fringes of cities, unlike in the main city and big towns. In most cities in sub-Saharan Africa, PUAs are fast becoming attractive to middle-class and higher income individuals whose lifestyle resembles those living in the city centre (Simon, 2008). This is because, land acquisition in PUAs is more affordable and less cumbersome. For instance, in Lagos city, many middle classes live in different PUAs such as Ikorodu in Lagos state, Akute, Agbado and Mowe in Ogun state. These middle-class people commute daily basis between these locations to Lagos for their daily activities and returned to their various homes. Furthermore, peri-urban lands are used for diverse activities and contain organized and disorganized cluster of residential, commercial and agricultural land uses [89]. Other significant forces contributing to the continued emergence of PUAs are mass migration towards large and primate cities in sub-Saharan African countries such Nigeria, Ghana, Cameroun, Togo, Mali, Burkina Faso, Kenya, Congo, Senegal etc. Also, the widening gap between population and economic

growth and the inability of surplus population in the city centre to emigrate to less populated countries are responsible for the increased spread of PUAs. In a nutshell, the major forces for converting and transforming land-use in PUAs can be categorized into two i.e., housing demand to meet the growing population [88] and the inadequate urban services and poor housing conditions [90]. Generally, PUAs are considered ugly development that usually leads to haphazard and disorganized layouts. It is typically seen as causing externality, which includes high dependence on automobile that always isolate with low income. Another explanation to advance the argument on the current trend of PUAs in sub-Saharan African cities is that higher economic gains accrue from using lands in PUAs for purposes aside agriculture. Fuseini & Kemp [91]; Fuseini et al., [92] argued that the lack of capacity by city authorities to provide adequate and affordable housing to meet the needs necessitated by the increasing growth rate of African cities led to the spread of PUAs. To mitigate or bridge this gap, many city authorities encouraged private sector participation in housing provision, contributing to the rapid expansion of most PUAs in sub-Saharan Africa. Therefore, this often results in uncontrolled and uncoordinated city expansion, eventually developing into PUAs otherwise known as urban sprawl. This rapid growth of cities leads to the encroachment of urban land use on rural land [93]. Tofowomo [94] also identified decentralization and density as conceptualizing PUAs. Decentralization means the spread of employment and population throughout the metropolitan area. At the same time, density refers to the level of jobs and population in high density living and working areas [94]. However, most popular views suggests that decentralization and low densities instead explain the emergence and spread of PUAs.

Several researches have provided explanations on the consequences of PUAs as having impacts on the environment [95], transportation and energy use [96, 97], public health [98] and socio-cultural changes [92, 99, 100]. In addition, PUAs could be said to be commonly found in developing countries and even worse, in sub-Saharan Africa. The spread of PUAs have negatively impacted on the environment due to the disruption of the ecosystem. The rapid reduction of green areas due to the uncontrolled felling of trees and clearing of vegetated areas for housing and other purposes have resulted to increased exposure to environmental hazards such as rise in temperature, poor air quality, erosion and flooding. Another consequence of PUAs is the inadequate access to market for goods and services couple with the deplorable condition of health, education, water, transportation and sanitation. PUAs also leads to unfavourable land-use pattern which hinders the development of sustainable transport modes, thereby increasing private car ownership that in turn results in increased numbers of trip generation and lengths, traffic congestion on the only main road linking the PUAs to the city centre. Due to the continued growth in the number of houses and businesses in the PUAs, local government authorities are forced to extend the provision of infrastructure which usually results in high cost and increased tax burden. Furthermore, PUAs also contributes to the segregation of rich and poor or what could be referred to as social isolation where low-income groups are abandoned in the core regions because they cannot afford transport fare to commute daily or car-based lifestyle. PUAs are also considered as agent of destruction to our natural resources [101], overuse and deterioration of soils and water resources [102]. Other environmental impacts of PUAs could be in form of water quality degradation, loss of wildlife and habitats, reduction in agricultural land use, destruction of natural landscape, and a shift in energy resource use. Also, the impacts of PUAs on quality of life include reduced physical activities, higher private vehicle ownership, higher trip generation and trip lengths, increase road fatalities, and greater health risk. Other serious negative effects of PUAs are destruction of unique natural areas, depletion of the ozone layer, climate change, degradation, and low interest in agricultural production.

However, PUAs have positive consequences. This includes the tendency to help reshape urban landscape [103–105]. In addition, PUAs help to control land prices in the city centre. They also make housing cheaper in the city centre because high-income earners filter to the suburbs. PUAs help reduce or prevent urban congestion and improve the quality of life. PUAs have positive ecological, environmental, social, economic, and community footprints but have opposing sides.

4.2.1 Case studies: Ghana and Nigeria

Urbanization is a significant driver of peri-urban growth [106]. It is an intense process that combines other processes to induce urban expansion. Hence, it has been described it as an amalgamation of processes that could be demographic, ecological, sociological, or economical [106]. This suggests that urbanization is multidimensional. Therefore, it is a salient force that could promote the occurrence of multiple intense activities, which could stimulate the expansion of urban areas into adjoining areas (PUAs). Concerning Africa, urban expansion is driven by local factors like natural population growth, changing land tenure system, expensive city rents, rural–urban migration, increase in demand for residential lands, and other related factors [107, 108]. However, consequential expansion does not bring about the much-needed socio-economic transformation required to boost human well-being and protect the environment because urbanization



Fig. 4 Eight Largest Urban Agglomerations in Nigeria. Source: UNDESA [85]

Table 4 Ten Major Ghanaian Cities

Rank	Metropolitan Area		Region
1	Accra	2,291,352	Greater Accra
2	Kumasi	2,069,350	Ashanti
3	Tamale	562,919	Northern Region
4	Sekondi-Takoradi	445,205	Western Region
5	Sunyani	248,496	Brong-Ahafo
6	Cape Coast	227,269	Central Region
7	Obuasi	180,334	Ashanti
8	Teshie	176,597	Greater Accra
9	Tema	161,612	Greater Accra
10	Koforidua	130,810	Eastern Region

Ghana Statistical Services [112]

process is largely unguided [109]. Therefore, rather than improve the situation at hand, it complicate issues by escalating poverty, reducing the prospect of socio-economic development and generally altering the functionality of urban areas [109]. The preceding implies that the pattern of urban expansion in Africa has hindrance sustainable development.

The preceding is true for cities in Nigeria and Ghana, where urbanization has induced expansion into the surrounding areas (PUAs) and caused the emergence of multiple informal settlements. For instance, Urbanet [110] predicted in 2015 that the eight most significant urban areas in Nigeria (which include Lagos, Ibadan, Benin City, Port Harcourt, Onitsha, Abuja, Kaduna, and Kano) will continue to grow bigger until 2030 with Lagos and three other cities are expected to increase by almost 200% (Fig. 4). Similarly, the urbanization process has induced urban sprawling into surrounding areas of major Ghanaian towns such as Accra, Kumasi, Tamale, Sekondi-Takoradi and other important cities in the country [111]. See Table 4 for other affected cities.

The Nigerian and Ghanaian PUAs share similarities and differences that cut across land use patterns to state of peri-urban infrastructures, pollution and waste management and others [108]. Most studies that could be cited to pinpoint the standard common points and disparities within the PUAs of the two countries deployed a merger of methodologies which were either quantitative (like remote sensing/GIS analysis) or qualitative (expert interview, and literature review) [108, 113, 114]. This is because urban expansion will not be easy to detect without the comparing of existing housing and other land uses in metropolitan areas with the emerging spatial land uses in the adjoining regions [108]. What this

suggests is that standard urban land use has more regularity and defined patterns due to better planning as against urban sprawl which may be characterized by mixed land uses, unevenness in building sizes, as well as scattered building [115, 133]

Table 5 is a collection of a few land use land cover change (LULCC) research works on PUAs in Ghana and Nigeria. The table revealed that the most everyday land uses land cover changes observable within the countries involve converting forest land use to other types of land uses i.e. agricultural, residential and non-residential (industrial) land uses. These were caused by factors that range from housing deficit to livelihoods, employment opportunities, ecosystem degradation, industrial development, natural population growth, rural–urban migration, highway and road development as well as policy change [37, 108, 113, 116–119].

The land use change across the two countries as shown in Table 4 are mostly similar, but snippets of disparities exist. For instance, Karg et al. [117] estimated the infrastructural index using their GIS analysis in their work on Tamale peri-urban areas and discovered that the study area enjoys quality infrastructural services span electricity, water, and good access to public transportation. This is not the case in Wa-Municipality and Obafemi-Owode PUAs, where Dapilah et al. [63], and Ayeni & Aborisade [37] reported that the areas lack access to basic amenities like viable road network and safe drinking water. Karg et al. [117, 119, 126, 134] testing the relationship between urbanization and infrastructure reported that a positive linear correlation exists between the variables. This can be interpreted to mean infrastructural availability is a pre-condition for urbanization which could be further interpreted to mean that poor infrastructural provision will negatively affect urbanization and by extension sustainable development.

Furthermore, Ikorodu and Ibeju-Lekki in the early 1980s were rural addendums of Lagos State, Nigeria. The areas were mainly used for agriculture, conservation, and water catchment purposes [113]. The predominant land uses in the regions were agriculture and residential in both areas but the tilt is that different factors induced these land uses were induced by different factors. Hence, the Ikorodu situation was caused by the housing deficit in core areas like Ikeja, Ibeju-Lekki's residential upsurge was the aftermath of highway development which linked the area to metropolitan areas like Victoria-Island [113]. Other dynamics typify SSA peri-urban areas, they include pollution, waste disposal, the land tenure system, and the activities of land speculators which are significant bottlenecks against sustainable development.

5 Findings and insights

Several factors have contributed to the rapid expansion of peri-urban areas in Nigeria. These factors, which range from physical to economic, social, and political, have influenced urban growth ([121–123, 136]. The dynamic and integrative nature of peri-urban areas has been a significant constraint, resulting in sprawled pattern development. The pattern of growth in the peri-urban area has been characterised by various undesirable changes in the fringe areas of cities, often consciously referred to as unplanned and uncontrolled development, resulting in poor and much unplanned suburbanization. Evidence from literature suggests that urban sprawl has positive and negative effect. Some researchers have identified urban sprawl as a source of urban and environmental problems. Several studies have identified and classified the following negative impacts of sprawl into economic, environmental, planning laws and policy and social dimension. The effect of urban sprawling is reviewed based on different sizes such as financial, planning laws and policy, environment, and social dimension. One of the main economic dimensions of the effect of sprawling is the severe shortage of affordable housing in areas where the sprawling is occurring [29]. Another effect of sprawling is the financial burden it imposes on the household as a result of mortgage debt [34, 35]. Individuals acquire house mortgages from financial institutions such as banks to develop their plots and/or buy plots, the mortgage is expected to be paid back according to the loan agreement payment terms, and this becomes a financial burden for individuals which may affect their take home pay and budget. In addition, it affects the residential developments, which largely contribute to sprawl. Also, increasing unemployment is one of the effects of sprawling [29]. Urban sprawl typically starts with population increase due to migration of people from different areas into a particular area. In most cases, jobs may not be readily available or insufficient to cater to people in a specific area. Transportation commuting issues are also a consequence of sprawling [29]. Effects of sprawl are that individuals who stay in sprawled areas are forced to use cars to commute to and from work/city centre/shopping/hospitals/ and those who do not own cars are forced to budget for transportation to and from their work place/city centre/shopping/hospitals etc.; this comes at a cost to individuals who have to budget for the cost of commuting either fuel for the car or transport money. Karwinska, Bohm, & Kudlacz, [30] identified the price of land impaired by

Table 5 Dynamics of Land use Change Works in Nigeria and Ghana

PUA Locations/Country	Type of land use and land cover change (LULCC)	Drivers of LULCC	Sources of information
Ikorodu (Nigeria)	Forests area converted to agriculture and residential landuse	Housing deficit in core areas, livelihood (agriculture)	Adedire [113]
Ibeju-Lekki (Nigeria)	Forests area converted to agriculture and residential landuse	Employment Opportunity, highway development	Adedire [113]
Odeda (Nigeria)	Forests area converted to agriculture and residential landuse	Livelihood (agriculture)	Tobore et al. [116]
Obafemi-Owode (Nigeria)	Forests area converted to residential and non-residential landuse	Ecosystem degradation (lumbering and land clearing)	Ayeni & Aborisade [37]
Tamale (Ghana)	Conversion of agricultural lands to residential and non-residential land uses	Industrial development, good accessibility (Roads)	Karg et al. [117]
Wa-Municipality (Ghana)	Forests area converted to agriculture and residential landuse	Industrial development (Sand & stone mining), Ecosystem degradation (Lumbering of Shea tree for charcoal production, bush burning by farmers)	Dapilah et al. [63]
Takoradi (Ghana)	Conversion of agricultural lands to residential and non-residential (industrial) land uses	Natural population growth, rural–urban migration, industrial development	Kleemann et al. [108]
Great Accra Metropolitan Area (GAMA) (Ghana)	Conversion of agricultural lands to residential	Policy change, rural–urban migration, road development	Agyeman [118]

urban sprawl falls (attractiveness) as one of the effects of sprawling. Sprawl lowers the attractiveness and value of the land. Christiawan [31] pointed out that sprawl areas depend primarily on imported food because the farmlands have been converted to residential area, and this reduces the amount of food that can be produced by farmers to feed their community and surplus to feed the nation, forcing, causing the importation of food. Higher infrastructure and public service costs, such as water supply, sanitation, electricity, public transport, waste management, policing and other services which become expensive to provide also affects the urban sprawl and reduced levels of public services [23, 29].

Planning Laws and Policy Dimensions affect urban sprawl in many ways. One of the ways is the lack of public transport, poorer connectivity, traffic congestion [23, 29]. In urban sprawling area, pressure is put on the existing public transport and road network due to the increased numbers of people in a sprawled locality, people use public transport or private cars which puts pressure on roads causing traffic congestion. Karwinska, Bohm & Kudlacz, [30] identified unplanned growth and un-coordinated development as a consequence of urban sprawling. Urban sprawling also causes an unequal distribution of public amenities [31]. The effects of sprawl are that providing services and infrastructural development to various locals is unequal. The aspect of the environmental dimension of the impact of urban sprawl is shown in the loss of vegetation (Habitat loss for fauna and flora or irreversible damage to local ecosystems), and biodiversity, habitat functions, agricultural resources, and soil are other form of environmental dimension of urban sprawling [23, 32]. Urban sprawl also causes increased (air, noise, light) pollution, water pollution and alteration of the hydrological properties of the water [23, 30, 31]. Energy and climate change (due to higher energy consumption and higher greenhouse gas emissions per individual) are consequences of urban sprawl [23].

The social Dimension aspect of urban sprawl concerns the social effects such as poverty, lack of opportunities, psychological problems, alcoholism, drugs, crime, violence and other irregular behaviors, limiting the quality of life, reduced community ties, missing cultural identity, loss of rural heritage [29, 31]. Different areas of the effect of urban sprawl are the higher costs for transport associated with commuting for families and increase in commuting time between homes, places of work and other activities [33]. People of various ethnicity, religions and income levels unable to live and work together because of segregation by income status. Also, urban sprawl causes increased car dependency, especially with weak or no connection to public transport; increased car dependency has adverse effects on health and accidents [34]. Public health issues such as stress, obesity, high blood pressure, respiratory problems (e.g. asthma due to air pollution) could also be as a result of urban sprawling [29, 34].

The issue of PUAs in sub-Saharan Africa has dramatically concerned many stakeholders, including city authorities, professionals in urban planning and management and scholars. Peri-urban area is a derivative of urbanization. It is seen as a way of escaping the problems associated with the city centres. This paper has examined the trends and consequences of PUAs in SSA, and give policy recommendations on how to plan sub-Saharan Africa's PUAs. The study found that the trends and consequences of PUAs in SSA exposes us to the current situation of PUAs and the consequences, which include both negative and positive. This paper identified some of the significant constraints of sustainable development in PUAs, which are categorized as follows: accessibility & infrastructural constraints, land use conversion, land tenure system, pollution/poor waste management, and an institutional debacles. The situation of PUAs in the sub-Saharan Africa indicates that peri-urban areas are fast becoming a visible occurrence in many sub-Saharan African cities and towns. For instance, the most major cities and towns in the sub-Saharan Africa have their fringes growing into a large-scale urban centre usually referred to as peri-urban areas that is as equally big as the size of the metropolitan area.

The negative consequences include the rapid reduction of green areas due to the uncontrolled felling of trees and clearing of vegetated regions for housing and other purposes which have increased exposure to environmental hazards such as rise in temperature, poor air quality, erosion and flooding. Others include inadequate access to the market for goods and services coupled with deplorable health, education, water, transportation, and sanitation conditions. PUAs also leads to unfavourable land-use pattern which hinder the development of sustainable transport modes, thereby increasing private car ownership which results in increased trip generation and lengths, and traffic congestion on the only main road linking the PUAs to the city centre. However, there are still positive side to PUAs. This includes the tendencies to help reshape the urban landscape. In addition, PUAs helps to control land prices in the city centre. It also makes housing to become cheaper in the city centre because of the filtering of high-income earners to the suburb. PUAs helps reduce or prevents urban congestion and improving the quality of life. PUAs have positive ecological, environmental, social, economic, and community foot prints but have opposing sides.

The expansion of PUAs in Nigeria and Ghana is not at the same level due to different population sizes. For instance, Nigeria has more urban centres than Ghana, which explains why the mass movement from urban to rural areas is higher in Nigeria than in Ghana. This further shows that PUAs spread faster in Nigeria than in Ghana, showcasing the considerable

differences in PUA expansion. As shown in Table 5, the dynamics of land use change in Nigeria and Ghana revealed that land use change in Nigeria and Ghana are mainly for residential and other purposes, which in this case bear some similarities.

6 Recommendation and implications

In conclusion, the paper recommends that government and city authorities must display the political will to develop a master-plan, which should contain a plan on effective urban management, effective control on the development on new buildings, effective urban governance, housing finance, implementation of master-plans, and provision of employment development. The government should muster the courage to implement the master-plan effectively. Furthermore, policy-makers are advised to suggest to the government to take over the lands in the urban fringes or by designated city managers to design a plan including the road networks and other specified purposes for the land. This will guide against the spread of disorganized and unplanned neighbourhoods currently experienced in both countries.

There have been several policies and suggestions on how to plan PUAs. However, this study will provide a more practical and straightforward remedy for the PUAs. Most governments or city managers have failed to manage and control of PUAs. The truth is that respective governments have largely neglected PUAs, and that is why PUAs in sub-Saharan Africa are not much better than slums. Respective city managers/authorities must prepare a policy document to manage the development of PUAs. The policy document should focus mainly on effective urban management, effective control of the development on new buildings, effective urban governance, implementation of master-plans, housing finance, provision of employment development and the political will to implement the policy effectively. Also, the government must show interest in the fringes of the cities where PUAs are found mainly by designing and implement a master plan for these regions. In addition, the government can also step up the provision, henceforth, city managers must make sure that rather than allow the city's fringes to sprawl, they must make sure that they develop a workable and implementable master plan for the development of the fringes in the town. Furthermore, governments must deliberate attempt to build infrastructure and encourage the location of industries by reducing taxes to provide employment opportunities for the residents of PUAs. This will reduce the economic gaps between PUAs and city centres. Also, the government can provide affordable, low-cost housing in different areas of the cities to discourage the drifts towards PUAs. The government should encourage the development of eco-city model in PUAs and city centres where residential areas are mixed with green space, parks and gardens, and trees and complemented with water fountains. A well-planned eco-city helps protect the environment from climate change and brings out the environment's beauty. Finally, all the suggestions above are to be aligned in a document. Above all, governments at all levels must muster the courage to implement the content of the policy to the letter.

7 Overall contribution

The study contributes to the literature by documenting differences and similarities in the spread of PUAs. It also contributes to knowledge by identifying and ascertaining the pattern of land use change in both Ghana and Nigeria, as shown in Table 5. The literature has also been enriched with discussions of the trends and consequences of the spread of PUAs.

8 Limitation of the study

The limitation encountered in this study is mainly in sourcing for the secondary data. The search for secondary data took a long time due to the many challenges of poor infrastructure in this part of the world. Each stage of the data search has been very stressful and time-consuming. There are numerous sources for secondary data, and getting the most appropriate one remains undaunting. Also, streamlining the numerous data or information obtained from these different sources was very tasking and time-demanding. Deciding on the most relevant among these data is very tasking.

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References

1. Salami AA, Tilakasiri SL, Ahamed YA. The indicators and indices of development. In: Tilakasiri SL, editor. *Geography in development: issues and perspectives*. Maharagama: Stamford Lake Pvt Ltd; 2017. p. 77–102.
2. Kutor SK. Development and underdevelopment of African continent: the blame game and the way forward. *Res Hum Soc Sci*. 2014;4(7):11.
3. Ikenna A. Understanding the crisis of development in Africa: reflections on Bedford Umez's analysis. *Afr J Political Sci Int Relations*. 2009;3(9):354–9.
4. Richard Peet EH. *Theories of development (revised)*. New York: Guilford Publications; 1999.
5. Todaro MP, Smith SC. *Economic development*. Italy: Addison-Wesley; 2012.
6. United Nations. 2018:WorldAtlas.com <https://www.worldatlas.com/regions/sub-saharan-africa.html>.
7. NEPAD. 2003. Action plan of the environment initiative of the new partnership for Africa's development. <http://www.economicswebinstitute.org/essays/nepad-environment.pdf>
8. Sloan S, Jenkins CN, Joppa LN, Gaveau DLA, Laurance WF. Remaining natural vegetation in the global biodiversity hotspots. *Biol Conserv*. 2014. <https://doi.org/10.1016/j.biocon.2014.05.027>.
9. Bhattacharyya S. Root causes of African underdevelopment. *J Afr Econ*. 2009. <https://doi.org/10.1093/jae/ejp009>.
10. United Nations Department of Economic and Social Affairs, Population Division. 2022. World population prospects 2022: Summary of results. UNDESA/POP/2022/TR/NO. 3.
11. United Nations Development Programme (UNDP). 2022. Human Development Report (HDR).
12. Gullette G, Thebpanya P, Singto S. Assessing urban expansion and livelihoods in Thailand's transitional spaces through combined ethnography and landsat data. *Hum Org*. 2017. <https://doi.org/10.17730/0018-7259.76.3.227>.
13. Oduro CY, Adamtey R, Ocloo K. Urban growth and livelihood transformations on the fringes of African cities: a case study of changing livelihoods in Peri-Urban Accra. *Environ Nat Res Res*. 2015. <https://doi.org/10.5539/enrr.v5n2p81>.
14. Ricci L. Peri-Urban livelihood and adaptive capacity: urban development in Dar Es Salaam. *Consilience J Sustain Dev*. 2012;7(1):71.
15. UNDESA. 2021. UNDESA World Social Report 2020. Department of economic and social affairs social inclusion.
16. Tiwari P. Dynamics of Peri Urban areas of Indian cities. *Int J Sci Eng Res*. 2019;10(4):40.
17. Camagni R, Gibelli MC, Rigamonti P. Urban mobility and urban form: the social and environmental costs of different patterns of urban expansion. *Ecol Econ*. 2002. [https://doi.org/10.1016/S0921-8009\(01\)00254-3](https://doi.org/10.1016/S0921-8009(01)00254-3).
18. Nilsson M, Lucas P, Yoshida T. Towards an integrated framework for sdgs: ultimate and enabling goals for the case of energy. *Sustainability*. 2013. <https://doi.org/10.3390/su5104124>.
19. Naab FZ, Dinye RD, Kasanga RK. Urbanisation and its impact on agricultural lands in growing cities in developing countries: a case study of Tamale in Ghana. *Modern Soc Sci J*. 2013;2(2):11.
20. OECD. 2018: Rethinking urban sprawl: moving towards sustainable cities, organisation for economic cooperation and development (OECD), www.oecd-ilibrary.org/sites/9789264189881-5-en/index.html?itemId=/content/component/9789264189881-5-en.
21. Gómez-Antonio M, Hortas-Rico M, Li L. The causes of urban sprawl in Spanish urban areas: a spatial approach. *Spat Econ Anal*. 2016;11(2):219–47.
22. Egidi G, Cividino S, Vinci S, Sateriano A, Salvia R. Towards local forms of sprawl: a brief reflection on mediterranean urbanization. *Sustainability*. 2020;12(2):582.
23. Aurambout JP, Barranco R, Lavalley C. Towards a simpler characterization of urban sprawl across urban areas in Europe. *Land*. 2018;7(1):33.

24. Pravitasari AE, Saizen I, Tsutsumida N, Rustiadi E, Pribadi DO. Local spatially dependent driving forces of urban expansion in an emerging Asian megacity: the case of Greater Jakarta (Jabodetabek). *J Sustain Dev*. 2015;8(1):108–19.
25. Makowska-Iskierka M. The role of the inhabitants of Łódź in the urbanisation of its surrounding area. *Turzym*. 2015;25(2):137–46.
26. Sitharam TG, Dhindaw J. Benefits and challenges of urbanization and low carbon energy needs in India. *J Sustain Urbaniz Planning Prog*. 2016;1(1):22–30.
27. McDonald RI. Implications of urbanization for conservation and biodiversity protection, encyclopedia of biodiversity. 2nd ed. Amsterdam: Academic Press; 2013.
28. Tagliarino NK, Bununu YA, Magbagbeola MO, De Maria M, Olusanmi AS. Paper submitted for the UN Economic Commission for Africa “2017 Conference on Land Policy in Africa” Nov. 14–17 2017, Addis Ababa, Ethiopia.
29. Jarah SHA, Zhou B, Abdullah RJ, Lu Y, Yu W. Urbanization and urban sprawl issues in city structure: a case of the Sulaymaniah Iraqi Kurdistan region. *Sustainability*. 2019;11(2):485.
30. Karwińska A, Böhm A, Kudłacz M. The phenomenon of urban sprawl in modern Poland: Causes, effects and remedies. *J Pub Gov*. 2018;45(3):26–43. <https://doi.org/10.15678/ZP.2018.45.3.02>.
31. Christiawan, P. I. (2019, November). Designing the mitigation model of urban sprawl potential impact in suburban Denpasar, Bali. In *Journal of Physics: Conference Series* (Vol. 1363, No. 1, p. 012100). IOP Publishing.
32. Gerten C, Fina S, Rusche K. The sprawling planet: simplifying the measurement of global urbanization trends. *Front Environ Sci*. 2019;7:140.
33. Frederick C, Riggs W, Gilderbloom JH. Commute mode diversity and public health: a multivariate analysis of 148 US cities. *Int J Sustain Trans*. 2018;12(1):1–11.
34. Chiguvu D, Kgathi-Thite D. Analysis of the positive and negative effects of urban sprawl and dwelling transformation in urban cities: case study of Tati siding village in Botswana. *J Legal Ethical Regul Isses*. 2022;25:1.
35. Lityński P. Living in sprawling areas: a cost–benefit analysis in Poland. *J Hous and the Built Environ*. 2023;38:1069–96. <https://doi.org/10.1007/s10901-022-09986-6>.
36. Mensah J, Ricart Casadevall S. Sustainable development: meaning, history, principles, pillars, and implications for human action: literature review. *Cogent Soc Sci*. 2019. <https://doi.org/10.1080/23311886.2019.1653531>.
37. Ayeni AO, Aborisade AG. An assessment of ecosystem degradation and poverty: a case study of Obafemi-Owode local government area, Ogun state, Nigeria. *Environ Urbaniz*. 2022. <https://doi.org/10.1177/09562478221080456>.
38. Cofie OO, Amerasinghe P, Larbi TO, Drechsel P. 2013. Facilitating outcomes: Multi-stakeholder processes for influencing policy change on urban agriculture in selected West African and South Asian cities. *IWMI Research Report*, 153. <https://doi.org/10.5337/2013.214>
39. Lawanson T, Yadua O, Salako I. 2012. Environmental challenges of peri-urban settlements in the Lagos megacity. In: Schrenk M, Popovich VV, Zeile P, Pietro EP (Eds). *Proceedings of Real Corp 2012: Re-Mixing the city—Towards sustainability and resilience?* Retrieved August 5, 2014, from http://www.corp.at/archive/CORP2012_40.pdf.
40. Fragkias M, Langanke T, Boone CG, Haase D, Marcotullio PJ, Munroe D, Olah B, Reenberg A, Seto KC, Simon D. 2012. Land teleconnections in an urbanizing world. In *Land teleconnections in an urbanizing world - GLP Report No. 5* (Issue 5).
41. United Nations International Children's Emergency Fund (UNICEF). 1999. *Urban Environmental Sanitation in Illegal Settlements*.
42. Pearce AR. 1999. The science and engineering of sustainability: a primer. <http://maven.gtri.gatech.edu/sfi/resources/pdf/TR/TR018.PDF>
43. Dyllick T, Hockerts K. Beyond the business case for corporate sustainability. *Bus Strat Environ*. 2002. <https://doi.org/10.1002/bse.323>.
44. UN. 2015. United Nations Sustainable Development Summit 2015. In *Transforming our world: the 2030 Agenda for Sustainable Development*.
45. Kates RW, Parris TM, Leiserowitz AA. What is sustainable development? Goals, indicators, values, and practice. *Environment*. 2005. <https://doi.org/10.1080/00139157.2005.10524444>.
46. Mensah AM, Castro LC. 2004. Sustainable resource use and Sustainable development: a contradiction?! *Desarrollo de Base : Revista de La Fundación Inter-americana*, 15(3).
47. Costanza R. The dynamics of the ecological footprint concept. *Ecol Econ*. 2000. [https://doi.org/10.1016/S0921-8009\(99\)00150-0](https://doi.org/10.1016/S0921-8009(99)00150-0).
48. Solow RM. On the intergenerational allocation of natural resources. *Econ Sustain*. 2017. <https://doi.org/10.4324/9781315240084-5>.
49. Cullen-Unsworth LC, Unsworth RKF. Strategies to enhance the resilience of the world's seagrass meadows. *J Appl Ecol*. 2016;53:967–72.
50. Pearce DW, Turner RK. *Economics of natural resources and the environment*, Baltimore MD: Johns Hopkins University Press, 1990, 378 pp., \$@@-@@42.50, \$@@-@@19.50 paper. *Am J Agric Econ*. <https://doi.org/10.2307/1242904>
51. Nahapiet J, Ghoshal S. Social capital, intellectual capital, and the organizational advantage. *Acad Manag Rev*. 1998;23(2):242–66. <https://doi.org/10.2307/259373>.
52. Wolff S, Mdemu MV, Lakes T. Defining the Peri-Urban: a multidimensional characterization of Spatio-temporal land use along an urban-rural gradient in Dar es Salaam. *Tanzania Land*. 2021;10(2):177. <https://doi.org/10.3390/land10020177>.
53. Smith MD, Sohail KM, Saywell DL. 1998. Basic Services in peri-urban Areas. In L. U. I. R. Columbia (Eds), *AGUA “98”: International Conference & Seminar*, CINARA, University of Cali.
54. Nuhu ADAAYIK. Characterisation of housing condition in Peri—Urban areas of Kaduna Metropolis, Nigeria. *Dutse J Pure Appl Sci*. 2021;7(1):317–25.
55. Ayeni AO, Aborisade AG, Onuminya TO, Soneye ASO, Ogundipe OT. Urban development in Africa and impact on biodiversity. *Curr Landscape Ecol Rep*. 2023. <https://doi.org/10.1007/s40823-023-00086-x>.
56. OECD. 1993. *Environmental performance reviews: a synthesis report by the group on the state of the environment*, environment monographs N° 83, organization for economic co-operation and development (OECD), [https://one.oecd.org/document/OCDE/GD\(93\)179/en/pdf](https://one.oecd.org/document/OCDE/GD(93)179/en/pdf).
57. Rey-Valette H, Laloë F, le Fur J. Introduction to the key issue concerning the use of sustainable development indicators. *Int J Sustain Dev*. 2007. <https://doi.org/10.1504/IJSD.2007.014410>.
58. Balestrat M. 2009. Spatial indicators for the analysis of peri-urban dynamics in the Languedoc Mediterranean area. *ERSA Territorial Cohesion of Europe and Integrative Planning*.

59. UNFPA. 2007. State of World Population 2007: Unleashing the Potential of Urban Growth. Director. United Nations Population Fund UNFPA
60. Bloch R, Fox S, Monroy J, Ojo A. 2015): Urbanisation and Urban Expansion in Nigeria. Urbanisation Research Nigeria (URN) Research Report. London: ICF International. Creative Commons Attribution-Non-Commercial-ShareAlike CC BY-NC-SA
61. Calderon C, Cantu C, Chuhan-Pole P. Infrastructure development in Sub-Saharan Africa: a scorecard. World Bank. 2018. <https://doi.org/10.1596/1813-9450-8425>.
62. Maryati S, Humaira ANS. 2015. Increasing the Infrastructure Access of Low-Income People in Peri-Urban of Bandung Metropolitan Area. *Int J Built Environ Sustain*. <https://doi.org/10.11113/ijbes.v2.n3.84>
63. Dapilah F, Nielsen JØ, Akongbangre JN. Peri-urban transformation and shared natural resources: the case of shea trees depletion and livelihood in Wa municipality. Northwestern Ghana *Afr Geograph Rev*. 2018. <https://doi.org/10.1080/19376812.2018.1480395>.
64. Ng A, Loosemore M. Risk allocation in the private provision of public infrastructure. *Int J Project Manage*. 2007. <https://doi.org/10.1016/j.jproman.2006.06.005>.
65. Infrastructure Consortium of Africa (ICA). 2018. Infrastructure financing trends in Africa.
66. AfDB. 2018. Africa's infrastructure: great potential but little impact on inclusive growth. In African economic outlook 2018.
67. Djeneba D, Lauridsen ML. 2019. Closing the SDG financing gap—trends and data, No. 73, <https://documents1.worldbank.org/curated/en/738131573041414269/pdf/Closing-the-SDG-Financing-Gap-Trends-and-Data.pdf>.
68. United Nations. 2014. World Urbanization Prospects: The 2014 Revision, United Nations Department of Economic and Social Affairs/Population Division. New York, NY, USA.
69. McConville J, Wittgren HB. (2014). Peri-Urban sanitation and water service provision challenges and opportunities for developing countries.
70. Appiah DO, Forkuo EK, Bugri JT. Land use conversion probabilities in a peri-urban district of Ghana. *Chin J Urban Environ Stud*. 2015;03(03):1550026. <https://doi.org/10.1142/S2345748115500268>.
71. Natural Water Retention Measures (NWRM). 2013. Land use conversion. <http://nwrn.eu/measure/land-use-conversion>
72. Nainggolan Ruth Roselin Erniwaty. Land use conversion: evaluation and strategic actions (case of Sumedang regency). *WIDYAPRAJA*. 2017;43(1):22–32.
73. Busko M, Szafranska B. Analysis of changes in land use patterns pursuant to the conversion of agricultural land to non-agricultural use in the context of the sustainable development of the malopolska region. *Sustainability*. 2018. <https://doi.org/10.3390/su10010136>.
74. Rawat US, Agarwal NK. Biodiversity: concepts, threats, conservation. *Environ Conserv J*. 2015;16(3):19.
75. Patra S, Sahoo S, Mishra P, Mahapatra SC. Impacts of urbanization on land use /cover changes and its probable implications on local climate and groundwater level. *J Urban Manage*. 2018. <https://doi.org/10.1016/j.jum.2018.04.006>.
76. Herrmann SM, Brandt M, Rasmussen K, et al. Accelerating land cover change in West Africa over four decades as population pressure increased. *Commun Earth Environ*. 2020;1:53. <https://doi.org/10.1038/s43247-020-00053-y>.
77. Asenso BB, Jia L, Menenti M, Yu L, Nyantakyi EK, Kabo-Bah AT, Jiang M, Zhou J, Lv Y, Zeng Y, et al. Spatiotemporal patterns in land use/land cover observed by fusion of multi-source fine-resolution data in West Africa. *Land*. 2023. <https://doi.org/10.3390/land12051032>.
78. World Bank. 2001. Making sustainable commitments—an environment strategy for the World Bank. <https://web.worldbank.org/archive/website00671/WEB/PDF/ENVIRONM.PDF>.
79. Neuwirth R. Shadow cities: a billion squatters, a new urban world. *Shadow Cities*. 2016. <https://doi.org/10.4324/9780203936245>.
80. Yang Y, Ye L. Peri-urban development. In *Urban Studies*. Oxford: Oxford University Press; 2020. <https://doi.org/10.1093/obo/9780190922481-0008>.
81. Thabit S, Aguinaga G, Maroso R, Mohn C, Edilbi B, Donnelly L, Berggren H. 2020. United Nations Human Settlements Programme (UN-Habitat).
82. UN-Habitat. 2014. State of African Cities 2014: re-imagining sustainable urban transitions, <https://unhabitat.org/state-of-african-cities-2014-re-imagining-sustainable-urban-transitions>.
83. Gomes SL. Interventions to strengthen institutional capacity for Peri-Urban water management in South Asia. *Water Sec Conflict Cooperation Peri-Urban South Asia*. 2022. https://doi.org/10.1007/978-3-030-79035-6_8.
84. Agheyisi JE. Land subdivision in peri-urban areas of Sub-Saharan African cities: conceptual definitions and policy guidelines. *Ghana J Geography*. 2018;10(1):9.
85. UN-DESA. 2014. United Nations Department of Economic and Social Affairs. 2015: World Urbanization Prospects: The Department of Economic and Social Affairs of the United Nations, Population Division. New York, NY, USA.
86. UNECA. 2017. An Overview of Urbanization and Structural Transformation in Africa. Economic Report on Africa 2017: United Nations Economic Commission for Africa: Addis Ababa, Ethiopia.
87. Appiah D, Bugri J, Forkuo E, Boateng P. Determinants of Peri-urbanization and land use change patterns in Peri-Urban Ghana. *J Sustain Dev*. 2014;7(6):95–109.
88. Acheampong RA, Anokye PA. Understanding households' residential location choice in Kumasi's Peri-Urban settlements and the implications for sustainable urban growth. *Res Humanit Soc Sci*. 2013;3(9):60–70.
89. Mandere MN, Ness B, Anderberg S. Peri-urban development, livelihood change and household income: a case study of peri-urban Nyahururu, Kenya. *J Agric Extension Rural Dev*. 2010;2(5):73–83.
90. Dutta V. Land use dynamics and peri-urban growth characteristics reflections on master plan and urban suitability from a sprawling North Indian city. *Environ Urbaniz ASIA*. 2012;3(2):277–301. <https://doi.org/10.1177/0975425312473226>.
91. Fuseini I, Kemp J. Characterizing urban growth in Tamale, Ghana: an analysis of urban governance response in infrastructure and service provision. *Habitat Int*. 2016;56:109–23.
92. Fuseini I, Yaro JA, Yiran GA. City profile: Tamale, Ghana. *Cities*. 2017;60:64–74.
93. Liu Y, Yue W, Fan P. Spatial determinants of urban land conversion in large Chinese cities: a case of Hangzhou. *Environ Plann B Plann Des*. 2011;38(4):706–25.
94. Tofowomo A. 2008. The planning implications of Urban Sprawl in Akure, 44th ISOCARP Congress.

95. Bart IL. Urban sprawl and climate change: a statistical exploration of cause and effect, with policy options for the EU. *Land Use Policy*. 2010;27(2):283–92.
96. Ahmadi L. 2014. The impact of sprawl on transportation energy consumption and transportation carbon footprint in large US cities. (Unpublished doctoral dissertation), University of Texas at Arlington, Texas, USA.
97. Rodríguez-Álvarez J. Urban energy index for buildings (UEIB): a new method to evaluate the effect of urban form on buildings' energy demand. *Landsc Urban Plan*. 2016;148:170–87.
98. Frumkin H. Urban sprawl and public health. *Public Health Rep*. 2002;117(3):201.
99. Hove M, Ngwerume E, Muchemwa C. The urban crisis in Sub-Saharan Africa: a threat to human security and sustainable development. *Stability Int J Sec Dev*. 2013;2(1):42.
100. Ji W, Ma J, Twibell RW, Underhill K. Characterizing urban sprawl using multi stage remote sensing images and landscape metrics. *Comput Environ Urban Syst*. 2006;30(6):861–79.
101. Price SJ, Dorcas ME, Gallantb AL, Klaverv RW, Willsonc JD. Three decades of urbanization: estimating the impact of land-cover change on stream salamander. *Biol Conserv*. 2006;133(4):436–41.
102. Haase D, Nuissl H. Does Urban sprawl drive changes in the water balance and policy? The case of Leipzig (Germany) 1870–2003. *Landsc Urban Plan*. 2007;80(1):1–13.
103. Elmes A, Mitchell CJA. Counter urbanites and commercial landscape change in the Canadian Countryside: Insights from Paris, Ontario. *J Rural Commun Dev*. 2020;15(1):49–70.
104. Downs A. Some realities about urban sprawl and urban decline. *Hous Policy Debate*. 1999;10(4):955–73.
105. Gordon P, Richardson H. Critiquing sprawl's critics. *Policy Anal*. 2000;365:1–18.
106. Cobbinah PB, Erdiaw-Kwasie MO. 2016. Urbanization in Ghana. In *E-planning and collaboration* (pp. 256–278). IGI Global. <https://doi.org/10.4018/978-1-5225-5646-6.ch012>
107. Ige VO, Ohiro EI. Factors contributing to peri-urban residential land prices near growing cities in developing countries. *Int J Econ Manage Stud*. 2020;7:198.
108. Kleemann J, Inkoom JN, Thiel M, Shankar S, Lautenbach S, Fürst C. Peri-urban land use pattern and its relation to land use planning in Ghana, West Africa. *Landscape Urban Planning*. 2017. <https://doi.org/10.1016/j.landurbplan.2017.02.004>.
109. Seifollahi-Aghmiuni S, Kalantari Z, Egidi G, Gaburova L, Salvati L. Urbanisation-driven land degradation and socioeconomic challenges in peri-urban areas: insights from Southern Europe. *Ambio*. 2022;51(6):1446–58. <https://doi.org/10.1007/s13280-022-01701-7>.
110. Urbanet. (2018). Infographics: urbanisation and urban development in Nigeria. Urbanet. <https://www.urbanet.info/urbanization-in-nigeria-infographics/>.
111. Owusu C. 2021. Urbanization in Ghana. Challenges and prospects for national development. GRIN. 23pages, <https://www.grin.com/document/1034782?lang=en>. Accessed on 10 Jul 2024.
112. Ghana Statistical Service (GSS), Ghana Health Service (GHS), and ICF International. 2015. Ghana Demographic and Health Survey 2014. Rockville, Maryland, USA: GSS, GHS, and ICF International. <https://www2.statsghana.gov.gh/docfiles/publications/2014%20GDHS%20%20Report.pdf>. Accessed on 10 Jul 2024.
113. Adedire F. 2021. Disparity in Peri-Urbanisation Process in Lagos, Nigeria. In *Landscape Architecture - Processes and Practices Towards Sustainable Development*. Intech Open. <https://doi.org/10.5772/intechopen.93530>
114. Bellwood-Howard I, Haring V, Karg H, Roessler R, Schlesinger J, Shakya M. 2015. Characteristics of urban and peri-urban agriculture in West Africa: results of an exploratory survey conducted in Tamale (Ghana) and Ouagadougou (Burkina Faso). <https://doi.org/10.5337/2015.214> Accessed 01 Jul 2024.
115. Chin N. 2002. Unearthing the roots of urban sprawl: a critical analysis of form, function and methodology. Working Papers Series—UCL Centre for Advanced Spatial Analysis, 47 (Mar 02). <https://discovery.ucl.ac.uk/id/eprint/249/> Accessed 22 Jul 2024.
116. Tobore A, Senjobi B, Oyerinde G. Spatio temporal analysis and simulation pattern of land use and land cover change in Odeda Peri-urban of Ogun state, Nigeria. *J Earth Environ Sci*. 2021;12(4):32.
117. Karg H, Hologa R, Schlesinger J, Drescher A, Kranjac-Berisavljevic G, Glaser R. Classifying and mapping peri-urban areas of rapidly growing medium-sized sub-Saharan African cities: a multi-method approach applied to tamale Ghana. *Land*. 2019. <https://doi.org/10.3390/land8030040>.
118. Agyeman SA. Impacts of urban sprawl on livelihoods and ecology in peri-urban fringe of the greater Accra metropolitan area. *West Afr J Appl Ecol*. 2018;26:1.
119. Parnell S, Pieterse E. 2014. eds. *Africa's Urban Revolution*. London: Zed Books, 2014, x + 309 pp.
120. Idowu O, Bako A, Aduloju O. Analysis of the trend of peri-urban development in Minna, Niger state. *J Geogr Inf Syst*. 2020;12:411–31. <https://doi.org/10.4236/jgis.2020.125025>.
121. Idowu, O.O. (2017) *Spatio-Temporal Analysis of Peri-urban Development in Minna, Niger State*. Ph.D. Thesis, University of Ilorin, Ilorin Kwara State.
122. Lamond, J., Awuah, B.K., Lewis, E., Bloch, R. and Falade B.J. (2015). *Urban Land, Planning and Governance Systems in Nigeria*. Urbanization Research Nigeria (URN) Research Report. ICF International, London. <https://assets.publishing.service.gov.uk/61250-URN> 1. Accessed 13 Jul 2024.
123. Noor NM, Rosini NA. Determination of spatial factors in measuring urban sprawl in Kuantan using remote sensing and GIS. *Procedia Soc Behav Sci*. 2013;85:502–12. <https://doi.org/10.1016/j.sbspro.2013.08.379>.
124. Butsch C, Heinkel SB. Peri-urban transformations in the Global South and their impact on water-based livelihoods. *Water*. 2020;12:458. <https://doi.org/10.3390/w12020458>.
125. Unwin, Tim (1989) "Urban-rural Interaction in Developing Countries: a Theoretical Perspective", in Potter and Unwin (eds.) *The Geography of Urban-rural Interaction in Developing Countries: Essays for Alan B. Mountjoy*, Routledge, London.
126. Tekalign GS. Urbanisation and rural development in sub-Saharan Africa: a review of pathways and impacts. *Res Globaliz*. 2023;6: 100133. <https://doi.org/10.1016/j.resglo.2023.100133>.

127. Sumardjo Firmansyah A, Dharmawan L. 2023. Peri-urban Community Resilience in Facing the Dynamics of Global Climate-change Impacts toward Sustainable Development. IOP Conference Series: Earth and Environmental Science, 1266, 3rd International Seminar of Natural Resources and Environmental Management.
128. Aboye BH, Gebre-Egziabher T, Kebede B. Peri-urban food insecurity and coping strategies among farm households in the face of rapid urbanization in Sub-Saharan Africa: evidence from Ethiopia. *Res Globaliz.* 2024;8: 100200.
129. Beyene F, Senapathy M, Bojago E, Tadiwos T. Rural household resilience to food insecurity and its determinants: Damot Pulasa district, Southern Ethiopia. *J Agric Food Res.* 2023;11:100500.
130. Cattaneo A, Adukia A, Brown DL, Christiaensen L, Evans DK, Haakenstad A, McMenomy T, Partridge M, Vas S, Weiss DJ. Economic and social development along the urban–rural continuum: New opportunities to inform policy. *Development review. World Dev.* 2022;157:105941.
131. Coronado-Apodaca KG, Martínez-Ruiz M, Iqbal HMN, Sosa-Hernandez JE, Parra-Saldívar R. Agro-food sustainability transitions: new frontiers for food security. *Curr Opin Environ Sci Health.* 2023;31:100412.
132. Dinku AM, Mekonnen TC, Adilu GS. Urban food systems: factors associated with food insecurity in the urban settings evidence from Dessie and Combolcha cities, north-central Ethiopia. *Heliyon.* 2023;9(3): e14482.
133. Ejaz N, Mallawaarachchi T. Disparities in economic achievement across the rural–urban divide in Pakistan: implications for development planning. *Econ Anal Policy.* 2023;77:487–512.
134. Chen M, Chen L, Cheng J, Yu J. (2022) Identifying interlinkages between urbanization and sustainable development goals. *Geography Sustain.* 2022;3(2022):339–46.
135. Dyachia ZS, Permana AS, Ho CS, Baba AN, Agboola OP. Implications of present land use plan on urban growth and environmental sustainability in a Sub-Saharan Africa city. *Int J Built Environ Sustain.* 2017. <https://doi.org/10.11113/ijbes.v4.n2.181>.
136. Fashagba JY, Ayeni AO. 2018. Natural resources: distribution, governance and politics. In Olorunfemi JO and SL Tilakasiri (Eds), *Human geography: concepts, approaches and trends.* (pp. 371–392). Stamford Lake (Pvt) Ltd.

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